Tyler Waltner

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EDUCATION

University of Rochester **Rochester, NY**

BS in Computer Science - Hajim School of Engineering *Graduation Date: May 2028*

BM in Jazz Saxophone Performance - Eastman School of Music *Graduation Date: May 2028*

Relevant Coursework: Deep Learning, Artificial Intelligence, Computer Optimizations, Data Structures and Algorithms

WORK EXPERIENCE

University of Rochester **Rochester, NY**

Undergraduate AI Security Researcher *Sep 2026 - Present*

- Engineered and replicated multimodal LLM safety experiments focused on jailbreaking methods (e.g., SEMA) using Python, producing 3+ reproducible evaluation pipelines across controlled epochs to measure attack success rates and mitigation impact.
- Analyzed model vulnerabilities across 3 distinct research phases, optimizing evaluation workflows to track attack success metrics and mitigation performance across 100+ iterative test runs.

Coding With Kids **Portland, OR**

Coding Instructor / Equipment Coordinator *Jun 2024 - Aug 2026*

- Taught interactive coding and STEM classes using specialized programming curricula, boosting student engagement and maintaining high classroom satisfaction ratings across sessions.
- Managed regional inventory and logistics as Equipment Coordinator for the PDX area, tracking hardware assets and orchestrating equipment exchanges across multiple locations to streamline company operations
- Optimized regional supply chains by executing hardware distribution and inventory audits, improving equipment availability and operational efficiency for local instructors.

Self Employed **Portland, OR**

Jazz Musician / Band Leader *Jun 2020 - Present*

- Managed business operations for 3 active jazz ensembles. (Overbridge, Tyler Waltner Quartet/Trio, Nu Zeta) Scaled the venture to $30K in yearly revenue through strategic client acquisition and venue bookings.
- Directed 100+ annual live performances across private events, weddings, and restaurants → coordinated rehearsal schedules and musical arrangements while maintaining a 100% event execution rate.

PROJECT EXPERIENCE

Dynamic CLI Minibatch MLP

Python, NumPy

- Engineered a custom Multi-Layer Perceptron from scratch using Python and NumPy without popular ML frameworks that classified the FashionMNIST dataset with custom He weight initialization and state caching.
- Scaled an original 1D-optimized SGD repository into a dynamic CLI-driven minibatch training pipeline that reduced execution overhead and improved hyperparameter configuration efficiency across training epochs.

NASA NeoW API Visualizer

React, Three.js, WebGL, HTML/CSS

- Built a 3D web application using React, Three.js, and WebGL that fetched and visualized near-Earth asteroid data from NASA's NeoW API.
- Optimized GPU memory management by bridging declarative React with imperative WebGL using useRef and smart caching that prevented API rate-limit violations and eliminated GPU memory leaks via a custom garbage-collection pipeline.

Map Routing & Pathfinding Visualizer

Java, Java Swing

- Engineered a geographic routing application in Java using Dijkstra's algorithm and a custom PriorityQueue that calculated the optimal shortest path between two locations.
- Designed a custom Graph data structure mapping intersections (vertices) and roads (edges) utilizing the Haversine formula that computed accurate real-world distances based on raw coordinate data.

- Developed a dual-interface system featuring a command-line parser and a Java Swing/Graphics2D GUI that visually rendered interactive maps and optimal navigation paths in real time.

TECHNICAL SKILLS

Coding Languages: Python, Java, C, JavaScript, HTML/CSS, R
Frameworks & Libraries: PyTorch, React, Node.js, JavaFX, NumPy, Matplotlib, Three.js, Pygame
Developer Tools: Git, Docker, VS Code, Visual Studio, Vim
Additional Skills: Bilingual (English/Japanese), Musical Instrumentation (Saxophones, Clarinet, Flute, Piano)